Complex Hamiltonian learning and KAM-preserving magnetic control of a catheter model

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Abstract

A learned model of a magnetic catheter should connect accurate prediction to bounded motion and realizable actuation. We develop this connection for a clamped–free beam in four steps. First, we derive its complex Hamiltonian representation and the complete fourth-order central symplectic update. Second, matched learning and integration comparisons quantify approximation and numerical errors. Third, nonzero-loss, state-only two-mode COMSOL identification yields coefficient and analytic-error bounds that support an a posteriori KAM proof for the fitted Hamiltonian at approximately 1.9 mm amplitude. Fourth, magnetic force-and-torque feedback realizes that conservative motion in the damped 64-mode plant. For zero higher-mode startup, the whole-centreline bound is below 1.971 mm. An input-work constraint additionally guarantees clearance during driven motion, including a quantified nonzero-error envelope over 4 s. Frequency continuation resolves a resonance-induced clearance boundary, and an S-shaped endpoint task achieves $6.419\,\mu$m RMS tracking error. Classical symplectic and KAM results underpin the analysis; the contributions are the model-specific error bounds, magnetic realization and full-model motion guarantees under stated sensing and actuation assumptions.

Keywords

magnetic catheter robots, complex Hamiltonian learning, symplectic integration, KAM theory, magnetic control

1 Background and research question

Magnetic actuation can steer a flexible catheter without placing motors along its body. A field applies torque to a magnetic segment, and a field gradient can supply force; elastic deformation and dissipation then determine the motion of the entire device (Abbott et al. 2020; Kim et al. 2019). For operation inside a confined passage, a useful dynamical model must therefore do more than predict the tip. It must connect available magnetic inputs to bounded centreline motion. This paper develops that connection for a clamped–free catheter model, using a learned Hamiltonian as the common object for numerical integration, nonlinear stability analysis and feedback design.

The application motivation is remote manipulation in reduced gravity. Spaceflight creates medical and operational conditions that warrant dedicated investigation; NASA's assessment of spaceflight-associated neuro-ocular syndrome is one example of a gravity-dependent health risk (NASA 2025). For a remotely manipulated device in this setting, we isolate the following mechanical question: how to sustain a prescribed class of bounded motions and respect a clearance limit when a flexible magnetic device has distributed inertia, elasticity and damping. The model is planar, has small slopes and neglects gravity and contact. These assumptions make the energy balance explicit and permit independent finite-element calculations, while identifying the physical effects that a later device study must supply.

Two operating objectives organize the analysis. The first is recurrent motion with a proved amplitude bound, represented by an invariant torus of a conservative model. The second is tracking a requested endpoint path while keeping the intervening body inside a corridor. The first objective asks whether motion persists for arbitrarily long times; the second asks how much external work can safely be supplied during a manoeuvre. A common mechanical energy links them, but they call for different conclusions. We use KAM theory for the recurrent motion and an input-work inequality for clearance, then derive magnetic commands that make those conclusions relevant to the same damped beam.

The main difficulty is preserving this connection through learning and actuation. Low prediction error alone leaves the Hamiltonian's nonlinear derivatives undetermined. Even

with accurate derivatives, a numerical solver can introduce an error comparable to the difference between two learned models. Finally, damping removes the recurrent motion of a conservative system unless a controller supplies the appropriate work. We address these issues in sequence: construct a power-consistent learned energy, resolve its approximation and integration errors, establish an invariant motion of the fitted function, and realize that motion with the available magnetic force and torque.

Recent magnetic-robot control studies address shape regulation and lumen navigation directly. Francescon et al. (2025) combine a pseudo-rigid-link model with vision feedback and homogeneous fields to control a robot with a prescribed lengthwise magnetization profile, demonstrating navigation in a soft pancreatic-duct phantom under varied insertion poses. Tong et al. (2025) use a geometrically nonlinear magnetic-rod simulation with reduced contact handling to compute fields that reduce distal-tip constraint forces, with planar phantom experiments and three-dimensional numerical demonstrations. These studies establish practical routes from a mechanical model to magnetic commands. The question pursued here is how a learned energy, with a quantified identification error, can support both persistent motion and bounds on the entire centreline. For the stated planar beam, we connect analytic error and twist bounds to a finite-amplitude invariant torus, realize its low-mode dynamics in the damped full finite-element plant, and bound driven motion through supplied work. Table 1 summarizes the objects passed between these steps.

1.1 Learning a mechanical model that can be analyzed

The starting point is a mechanical energy with explicit boundary work. Variational beam and rod formulations relate distributed strain energy, inertia and applied loads before discretization (Simo 1985; Antman 2005). In the present specialization, the tip magnet contributes a cosine potential under a static bias field. Its curvature enters the linear modal frequencies, while its higher-order terms govern the amplitude dependence of those frequencies. We retain the complete cosine because the nonlinear part later supplies the twist required for KAM nondegeneracy. Modal coordinates are normalized once, and the same dimensional map is used to interpret learned coefficients, simulated trajectories and magnetic commands. Supplement S1 supplies the distributed derivation; Section 2 develops the model used in the analysis.

Hamiltonian neural networks provide a natural learning mechanism: fit a scalar energy and obtain its canonical vector field by differentiation (Greydanus et al. 2019). This makes the energy derivatives available to both numerical integration and stability analysis. A magnetic catheter also loses and receives energy, so we follow the separation of conservative dynamics, dissipation and inputs used by dissipative SymODEN and port-Hamiltonian networks (Zhong et al. 2020; Desai et al. 2021). The learned conservative part specifies reversible dynamics; a nonnegative dissipation model and explicit magnetic ports complete the physical decoder. This separation is necessary for a controller to distinguish the energy stored by the device from the work it must supply.

Complex coordinates package modal position and momentum into one state. This representation supports shared complex features (Trabelsi et al. 2018). Their physical interpretation remains tied to the underlying real energy. We derive that representation explicitly, including the conjugate dependence and the analytic extension used in the KAM argument. Complex-network approximation theory offers useful existence and comparison results, but those results depend on the activation, target class and norm (Zhang et al. 2022). Here the relevant question is whether the fitted model controls the required derivatives on a declared domain. Smooth approximation theory (Hornik 1991) provides an existence route; the quantitative analysis then uses the actual finite feature class and its fitted coefficients.

That feature class combines the complete magnetic potential with an analytic polynomial residual. Its coefficients can be fitted from energy derivatives or state increments, and all derivatives needed for the subsequent analysis are explicit. A residual may depend on momentum as well as position. We retain that dependence because it changes both the Hamiltonian vector field and the relation between physical velocity and canonical momentum. The magnetic feedback is consequently derived from the full learned gradient and Hessian. This gives learning a concrete role: it supplies a differentiable energy whose approximation error can be bounded and whose dynamics can be translated back into physical commands.

The distinction between identification and model reduction is particularly important for this device. A low-mode projection of a full beam trajectory need not satisfy a closed low-mode equation: the discarded modes also contribute to tip rotation and hence magnetic torque. A fitted residual can absorb part of that discrepancy on the observed trajectories. Its predictive value should therefore be assessed against independent full-beam solutions, while a theorem about that

Table 1. Four steps of the study. Each step supplies the object needed by the next; the detailed proofs and additional experiments are in the Supplement.

Step	Question	Result passed to the next step
1. Derive, Section 2	Which energy and discrete update describe the beam?	A power-consistent complex representation and a complete central symplectic update.
2. Evaluate, Section 3	What errors remain after fitting and integration?	Paired comparisons and quantitative approximation bounds for a specified learned function.
3. Establish persistence, Section 4	Does that function preserve a nonlinear recurrent motion?	Nondegenerate twist, a finite-amplitude invariant torus and a local family of tori.
4. Realize, Section 5	How does the motion guide magnetic actuation?	Force/torque feedback, full-model amplitude bounds and clearance-constrained tracking.

residual must use the function actually obtained by fitting. We organize the computations around both questions. Controlled reduced-model data isolate coefficient recovery and permit a quantitative learning-to-KAM argument. Independent full finite-element observations assess the discrepancy introduced by reduction and the response to driven motion.

The comparison design separates the effects of representation and mechanical knowledge. Real and complex implementations are fitted with and without the known quadratic energy, using paired training seeds and the same held-out trajectories. Five paired contrasts measure the effect of the prior in each representation, the representation effect with and without that prior, and their interaction. A separate integration comparison resolves numerical error before small learning differences are interpreted. The resulting real and complex predictions are equivalent at the resolved precision for this feature class.

1.2 *From numerical trajectories to persistent physical motion*

A learned Hamiltonian is continuous in time, whereas prediction and control require discrete updates. Central generating functions provide classical symplectic difference schemes for this step (Feng et al. 1989). We derive the fourth-order central formula through its midpoint expansion and specialize it to the fitted magnetic energy. The correction, its symplecticity and its order belong to the established geometric-integration framework (Hairer et al. 2006); the model-specific work is the complete realization for the learned, generally nonseparable Hamiltonian and its numerical evaluation. This treatment exposes the derivatives actually required by the update and allows the solver tolerance, phase error and state error to be assessed together.

The resulting numerical trajectories motivate the stability question, but KAM persistence is determined by properties of the Hamiltonian itself. Classical KAM theory connects sufficiently small perturbations of nondegenerate integrable systems to families of invariant tori (Pöschel 2001; Bounemoura 2020). For the magnetic beam, the nonlinear frequency variation can be computed from the retained cosine. Learning must preserve that variation and control the full remainder. The analysis therefore follows coefficients and analytic norms through to the theorem assumptions.

Chen et al. connect nonzero Hamiltonian-network training loss to KAM theory under statistical and regularity hypotheses (Chen et al. 2022). We develop a computable counterpart for the present analytic feature class. In the experiment used for the finite-amplitude proof, state-only COMSOL observations determine an integral regression. Bounds on its residual, integration defect and design matrix yield a coefficient-error bound; differentiating the same features converts it into analytic-error and twist bounds. Thus the learning error remains part of the argument even though the training loss is nonzero. The broader statistical route is retained in Supplement S3, while the main proof uses the deterministic bounds appropriate to these trajectory observations.

With those bounds in place, an a posteriori KAM theorem tests a computed torus of the fitted Hamiltonian through its invariance defect and geometric conditions (Haro and Luque 2019). We apply it to the exported learned function, including its complete cosine and momentum-dependent residual. The result is a torus at approximately 1.9 mm amplitude. Nondegeneracy of its final torsion also supplies a local family of positive measure through an established torus-stability result (Bounemoura 2016). Explicit measure estimates require a specified domain and constants (Biasco and Chierchia 2018; Chierchia and Koudjinan 2021); we retain that calculation in the Supplement. Weighted Birkhoff averages additionally explore a wider domain over finite times (Das et al. 2017; Duignan and Meiss 2023), providing a complementary numerical view of the recurrent motions.

For these motions to guide actuation, the conservative subsystem must be connected to the damped full beam. We use a transverse magnetic force and torque to reproduce the learned low-mode accelerations. The remaining modes

then form stable forced oscillators, whose quasiperiodic responses can be constructed and bounded. This yields an invariant torus of the complete 64-mode model and a bound that also includes startup from zero higher-mode states. Leaving the torus for an endpoint task invokes a second control mechanism: a magnetic-work governor limits physical energy and hence the whole-centreline deflection. The two mechanisms address recurrent operation and manoeuvring within the same mechanical setting.

1.3 Contributions and study design

The work contributes three model-specific results. First, it derives the complex magnetic Hamiltonian, implements the classical fourth-order central update for the learned nonseparable energy, and resolves representation, prior and integration effects through matched comparisons. Second, it converts nonzero state-only identification error into computable analytic and twist bounds, then verifies a finite-amplitude invariant torus of that fitted function. Third, it constructs magnetic feedback and a full-model torus lift, together with an energy-based clearance condition for driven motion. The general approximation and KAM theorems are established results, cited with their hypotheses and identifiers where applied; the coefficient bounds, mechanical constructions and application proofs are developed here.

The quantitative results make the connection operational within the stated model. Starting the low state on the certified torus and all higher states at zero gives a whole-centreline bound below 1.971 mm, a transverse-field bound below 0.922 mT and a gradient bound below 0.02573 T/m. A separate S-shaped endpoint task achieves $6.419\,\mu$m RMS error with the learned controller.

Two COMSOL identification settings serve distinct purposes within this common mechanical model: full finite-element observations test prediction and tracking, whereas the two-mode integral observations supply the learning-to-KAM proof and the associated magnetic realization. Their fitted coefficients and error budgets are specified separately. Section 6 brings the results together.

2 Derivation of the complex Hamiltonian and symplectic scheme

2.1 From magnetic rod mechanics to canonical variables

We first derive an energy from magnetic beam mechanics, express that energy in complex coordinates, and then discretize its conservative flow. For a planar, small-slope cantilever, the transverse displacement $w(s,t)$ satisfies

$$\begin{aligned} \rho A_c w_{tt} + c_d w_t + EI w_{ssss} &= 0, \qquad 0 < s < L, \\ w(0,t) = w_s(0,t) &= 0, \qquad w_{sss}(L,t) = 0, \\ EI w_{ss}(L,t) &= \mu[B_y \cos\vartheta - B_0 \sin\vartheta], \end{aligned} \tag{1}$$

where $\vartheta = w_s(L,t)$, $A_c = \pi r_c^2$, and $I = \pi r_c^4/4$. The negative rotation derivative of the magnetic potential $-\mu(B_0\cos\vartheta + B_y\sin\vartheta)$ gives the tip moment. Thus the free-end rotation remains unconstrained, the constant bias B_0 stores energy, and the transverse field $B_y(t)$ supplies work.

The 80 mm beam has radius 0.6 mm and uses 32 consistent-mass cubic-Hermite elements, leaving 64 free coordinates after clamping; supplementary Table S1 gives its material and field parameters. Let q, M, K and e_ϑ denote the nodal vector, mass matrix, bending stiffness and tip-rotation selector. Two integrations by parts in the Galerkin weak form place the natural moment on e_ϑ, giving the finite-element equations and stored energy

$$\begin{aligned} M\ddot{q} + \alpha M\dot{q} + Kq &= \mu e_\vartheta[B_y\cos\vartheta - B_0\sin\vartheta], \\ H_0(q,p) &= \tfrac{1}{2}p^{\mathsf{T}}M^{-1}p + \tfrac{1}{2}q^{\mathsf{T}}Kq \\ &\quad + E_0[1-\cos(e_\vartheta^{\mathsf{T}}q)], \end{aligned} \tag{2}$$

where $p = M\dot{q}$, $E_0 = \mu B_0 = 6.0\times10^{-7}$ J and $c_d = \alpha\rho A_c$, with $\alpha = 0.55558524\,\mathrm{s}^{-1}$. The cosine potential remains exact within the small-slope beam model.

The bias-field stiffness is $K_b = K + E_0 e_\vartheta e_\vartheta^{\mathsf{T}}$. Choose $\Phi_n^{\mathsf{T}} M \Phi_n = I$ and $\Phi_n^{\mathsf{T}} K_b \Phi_n = \mathrm{diag}(\omega_j^2)$ for n retained modes, and write $q = \Phi_n a$, with reduced canonical momentum $\pi = \dot{a}$. Set $\omega_0 = \omega_1$, $\tau = \omega_0 t$, and $\nu_j = \omega_j/\omega_0$. Define the scalar position scales $S_j = \sqrt{E_0/(\omega_0\omega_j)}$. The balanced scaling is

$$a_j = \sqrt{\frac{E_0}{\omega_0\omega_j}}X_j, \qquad \pi_j = \sqrt{\frac{E_0\omega_j}{\omega_0}}Y_j. \tag{3}$$

The scale product E_0/ω_0, energy H_0/E_0 and time $\tau = \omega_0 t$ give canonical dimensionless equations. Define $S = \Phi_n\,\mathrm{diag}(S_j)$, $d = S^{\mathsf{T}}e_\vartheta$, and $\ell(s)^{\mathsf{T}} = N(s)S$, where $N(s)$ is the Hermite displacement row. Then $w(s) = \ell(s)^{\mathsf{T}}X$, $\vartheta = d^{\mathsf{T}}X$, and

$$\begin{aligned} h_*(X,Y) &= h_2(X,Y) + g(d^{\mathsf{T}}X), \\ h_2 &= \tfrac{1}{2}\sum_{j=1}^{n}\nu_j(X_j^2+Y_j^2), \\ g(s) &= 1-\cos s - \tfrac{1}{2}s^2. \end{aligned} \tag{4}$$

Subtracting $s^2/2$ avoids counting the bias stiffness twice. For the two-mode learning model, $\omega_0 \approx 11.417287283\,\mathrm{rad\,s^{-1}}$, $\nu \approx (1, 6.152117504)$, $d \approx (0.225670773, 0.321798806)^{\mathsf{T}}$, and $\ell(L) \approx (0.013506054, 0.005470926)^{\mathsf{T}}\,\mathrm{m}$. Calculations use the full exported coefficients in this fixed chart, shared by learning, integration and stability analysis.

2.2 Complex energy and a power-consistent decoder

The complex Hamiltonian follows by an invertible change of coordinates. Let $x = (X, Y) \in \mathbb{R}^{2n}$, $J = \left(\begin{smallmatrix} 0 & I \\ -I & 0 \end{smallmatrix}\right)$, and

$$\zeta_j = \frac{X_j + \mathrm{i}Y_j}{\sqrt{2}}, \qquad \chi_j = \frac{X_j - \mathrm{i}Y_j}{\sqrt{2}},$$
$$X_j = \frac{\zeta_j + \chi_j}{\sqrt{2}}, \qquad Y_j = \frac{\zeta_j - \chi_j}{\mathrm{i}\sqrt{2}}. \tag{5}$$

On the real phase space $\chi = \bar{\zeta}$. Substitution into (4) gives the explicit complex form

$$h_*^{\mathbb{C}}(\zeta, \chi) = \sum_j \nu_j \zeta_j \chi_j + 1 - \cos\left(\frac{d^{\mathsf{T}}(\zeta + \chi)}{\sqrt{2}}\right) - \tfrac{1}{4}[d^{\mathsf{T}}(\zeta + \chi)]^2. \tag{6}$$

It restricts to the real energy when $\chi = \bar{\zeta}$ and is entire in the independent complex variables (ζ, χ). This extension, including the learned polynomial residual below, supplies the analytic Hamiltonian used in KAM theory. It need not be holomorphic in ζ alone. For a real analytic energy h, the chain rule yields

$$\{\zeta_j, \bar{\zeta}_k\} = -\mathrm{i}\delta_{jk}, \qquad \partial_{\bar{\zeta}_j} = \frac{\partial_{X_j} + \mathrm{i}\partial_{Y_j}}{\sqrt{2}},$$
$$\zeta' = -\mathrm{i}\partial_\chi h^{\mathbb{C}}, \qquad \chi' = \mathrm{i}\partial_\zeta h^{\mathbb{C}}. \tag{7}$$

Magnetic actuation and damping enter through the physical decoder. A CHNN supplies real readouts h_η and $\mathcal{D}_\eta(X, v)$, with physical dissipation divided by $E_0\omega_0$, and

$$X' = v = \partial_Y h_\eta,$$
$$Y' = -\partial_X h_\eta - \partial_v \mathcal{D}_\eta(X, v) + u\, d\cos(d^{\mathsf{T}}X). \tag{8}$$

Here $u = B_y/B_0$, and $\partial_v \mathcal{D}_\eta$ is evaluated before substituting $v = \partial_Y h_\eta$. With $\mathsf{N} = \mathrm{diag}(\nu_j)$, we use $\mathcal{D}_\eta = \widehat{\gamma} v^{\mathsf{T}} \mathsf{N}^{-1} v/2$, $\widehat{\gamma} \geq 0$; for the stated beam, $\gamma = \alpha/\omega_0$. Taking the derivative along the decoded trajectory gives

$$h_\eta' = -v^{\mathsf{T}} \partial_v \mathcal{D}_\eta + u\cos(d^{\mathsf{T}}X) d^{\mathsf{T}} v, \tag{9}$$

with an additional $\partial_\tau h_\eta$ for time-varying parameters. The quadratic Rayleigh readout makes the first term nonpositive. In complex coordinates this decoder becomes

$$\zeta' = -\mathrm{i}\partial_{\bar{\zeta}} h_\eta + \frac{\mathrm{i}}{\sqrt{2}}[-\partial_v \mathcal{D}_\eta + u\, d\cos(d^{\mathsf{T}}X)]. \tag{10}$$

One may package the readouts as $\mathcal{H}_\eta = h_\eta + \mathrm{i}\mathcal{D}_\eta$, but this pair is distinct from the analytic continuation $h_\eta^{\mathbb{C}}$: its dynamics are defined by the power-consistent decoder above. The symplectic scheme acts on the conservative energy with $u = 0$ and $\mathcal{D}_\eta = 0$.

2.3 Derivation of the fourth-order central scheme

For $h = h_\eta$ and $f = J\nabla h$, we derive the specialization of Feng et al.'s classical fourth-order central scheme (Feng et al. 1989, Theorem 24, p. 90; fourth-order formula, p. 95) to the full learned magnetic energy. The generating-function construction (Hairer et al. 2006) retains all mixed position–momentum terms. Let Δ denote the dimensionless step and $m = (x_- + x_+)/2$ the endpoint midpoint. A central generating function $\mathcal{S}_\Delta$ defines a canonical map through

$$x_+ - x_- = J\nabla \mathcal{S}_\Delta(m). \tag{11}$$

Taking $\mathcal{S}_\Delta = \Delta h$ gives implicit midpoint. To determine the cubic correction, expand the exact trajectory about its temporal midpoint $x_0 = x(t + \Delta/2)$:

$$m = x_0 + \tfrac{1}{8}\Delta^2 f' f(x_0) + O(\Delta^4),$$
$$x_+ - x_- = \Delta f(x_0) + \tfrac{1}{24}\Delta^3\{f''[f, f] + (f')^2 f\}(x_0) + O(\Delta^5). \tag{12}$$

Since $x_0 = m - \Delta^2 f' f(m)/8 + O(\Delta^4)$, substitution yields

$$x_+ - x_- = \Delta f(m) + \frac{\Delta^3}{24}\{f''[f, f] - 2(f')^2 f\}(m) + O(\Delta^5). \tag{13}$$

The coefficient $-2/24 = 1/24 - 1/8$ accounts for the difference between the temporal and endpoint midpoints.

To recover a scalar generator, put $A = \nabla^2 h$. Since $f' = JA$ and $(f')^{\mathsf{T}} = -AJ$, differentiation gives

$$\nabla(f^{\mathsf{T}} A f) = h^{(3)}[f, f] - 2AJAf,$$
$$J\nabla(f^{\mathsf{T}} A f) = f''[f, f] - 2(f')^2 f. \tag{14}$$

Consequently choose

$$\mathcal{S}_\Delta(x) = \Delta h(x) + \frac{\Delta^3}{24} f(x)^\mathsf{T} \nabla^2 h(x) f(x). \tag{15}$$

Equations (11) and (14) give the fourth-order difference scheme

$$\begin{aligned} x_+ &= x_- + g_\Delta(m), \\ g_\Delta &= \Delta f + \frac{\Delta^3}{24}[f''[f,f] - 2(f')^2 f]. \end{aligned} \tag{16}$$

Equivalently, $\zeta_+ - \zeta_- = -\mathrm{i}\partial_{\bar{\zeta}} \mathcal{S}_\Delta(m)$.

Each step solves $r(x_+) = x_+ - x_- - g_\Delta(m) = 0$. Newton's correction is

$$x_+^{(k+1)} = x_+^{(k)} - \left[I - \tfrac{1}{2} D g_\Delta(m^{(k)})\right]^{-1} r(x_+^{(k)}). \tag{17}$$

The Jacobian uses energy derivatives through order four and is invertible near the identity for sufficiently small Δ. Analytic polynomial and cosine derivatives therefore provide an implementable scheme even when learned momentum terms preclude the physical kinetic–potential split.

3 Numerical and approximation analysis

3.1 Symplecticity, order and phase accuracy

For a differentiable local solution of (16), set $C_\Delta = J\mathcal{S}''_\Delta(m)/2$. Differentiating the implicit map gives

$$D\Psi_\Delta = (I - C_\Delta)^{-1}(I + C_\Delta). \tag{18}$$

Since $\mathcal{S}''_\Delta$ is symmetric, $C_\Delta^\mathsf{T} J + J C_\Delta = 0$, so this Cayley transform satisfies $(D\Psi_\Delta)^\mathsf{T} J D\Psi_\Delta = J$. The odd dependence $\mathcal{S}_{-\Delta} = -\mathcal{S}_\Delta$ also gives $\Psi_{-\Delta} = \Psi_\Delta^{-1}$. The local defect $O(\Delta^5)$ in (13) implies fourth-order convergence on fixed intervals with bounded derivatives and locally unique implicit steps. These classical properties (Feng et al. 1989, Theorems 17–18 and 23–24) hold for the complete fitted energy; the computation below measures the additional error from finite Newton solves.

The linear oscillator makes the phase improvement explicit. For $h = \nu(X^2 + Y^2)/2$, put $a = \nu\Delta + (\nu\Delta)^3/12$. The step is a rotation with eigenvalues $(1 \pm \mathrm{i}a/2)/(1 \mp \mathrm{i}a/2)$ and phase magnitude

$$\theta_\Delta = 2\arctan(a/2) = \nu\Delta - \frac{(\nu\Delta)^5}{120} + O(\Delta^7). \tag{19}$$

Its modulus is one for every real step. Implicit midpoint instead has $\theta_\Delta = \nu\Delta - (\nu\Delta)^3/12 + O(\Delta^5)$. Both preserve the quadratic energy exactly while accumulating different phase errors. For a nonlinear energy, symplecticity preserves canonical structure and does not imply exact energy conservation.

Table 2. Endpoint errors for the complete fitted GC17 energy. The state error is Euclidean; orders use successive step halvings.

Δ	Midpoint error (order)	Central4 error (order)
0.08	7.1351×10^{-2} (—)	1.8346×10^{-3} (—)
0.04	1.9383×10^{-2} (1.8802)	1.1784×10^{-4} (3.9606)
0.02	4.8966×10^{-3} (1.9849)	7.4149×10^{-6} (3.9902)
0.01	1.2265×10^{-3} (1.9972)	4.6422×10^{-7} (3.9976)

We test this distinction on the prior-informed complex fit from training seed 17 (GC17, defined in Section 3.3), retaining its complete cosine and quartic terms. From initial state $(0.1, 0.06, 0, 0.02)$ we integrate to $\tau = 10$ against a DOP853 reference whose tolerance refinement changes each component by at most 8.32×10^{-14}. Table 2 confirms the predicted orders.

The maximum implicit residual is 2.00×10^{-15} and the maximum symplectic defects are 3.34×10^{-16} for Central4 and 2.24×10^{-16} for midpoint. An independent finite-difference tangent agrees within 1.39×10^{-12}. These results confirm fourth-order convergence and structure preservation at the measured solve accuracy. The comparison uses equal step sizes; Central4 requires higher derivatives. Supplementary Section S9.1 gives the solver tolerances and complete checks.

3.2 Complex features and derivative approximation

Having controlled integration error, we next control the learned energy and its derivatives. The complex form (6) suggests retaining the magnetic cosine and learning a quartic residual. For two modes, let $m_4(x)$ list the 35 degree-four monomials, v_4 contain the coefficients of $-(d^\mathsf{T} X)^4/24$, and the 34 columns of U span its orthogonal complement. Define

$$\begin{aligned} F(x) &= \begin{pmatrix} g(d^\mathsf{T} X)/\|v_4\|_2 \\ U^\mathsf{T} m_4(x) \end{pmatrix}, \\ h_\eta(x) &= h_2(x) + \eta^\mathsf{T} F(x) = h_2 + a\, g(d^\mathsf{T} X) + m_4(x)^\mathsf{T} U c, \end{aligned} \tag{20}$$

where $\eta = (a\|v_4\|_2, c)$. This class contains the target at $a = 1, c = 0$ without duplicating its leading quartic direction. Substituting (5) realizes each monomial with complex affine forms and their conjugates; square activations implement products via $uv = [(u+v)^2 - (u-v)^2]/4$. The cosine and these polynomial features give a finite analytic complex

network with a real linear readout and allow momentum-dependent residuals.

More generally, increasing the polynomial degree approximates an analytic target and each fixed order of its derivatives on a compact polydisk inside the power-series convergence domain. The fixed class above instead admits a quantitative coefficient bound. This derivative requirement differs from the L^2 comparison for the particular function classes in Zhang et al. (2022, Theorem 2).

For an in-class target $h_* = h_2 + \eta_*^{\mathsf{T}} F$, define $B_r(D) = \max_{|\beta|\le r} \sup_{x\in D} \|\partial^\beta F(x)\|_2$. For conservative derivative labels, let ℓ be the mean over N observations of the squared Euclidean residual of each four-component vector, and ϵ_{rms} the corresponding label-error bound. With $DF \in \mathbb{R}^{35\times 4}$ and $G = N^{-1}\sum_i DF(x_i)DF(x_i)^{\mathsf{T}}$, the triangle inequality gives

$$\begin{aligned} \|h_\eta - h_*\|_{C^r(D)} &\le B_r(D)\|\eta - \eta_*\|_2 \\ &\le \frac{B_r(D)(\sqrt{\ell} + \epsilon_{\mathrm{rms}})}{\sqrt{\lambda_{\min}(G)}}, \end{aligned} \tag{21}$$

provided G is positive definite; the C^r norm is the maximum over scalar coordinate derivatives. Indeed, orthogonality of J equates the vector-field coefficient error to the derivative-feature error, and the least eigenvalue of G bounds its empirical squared norm from below. This model-specific estimate includes nonzero fitting and label errors. A target outside the class requires an additional approximation defect in both the label budget and function error. Section 4 uses these quantities to test KAM conditions.

3.3 Matched identification and long-time prediction

Five paired comparisons separate representation from the mechanical prior. R and C denote real and complex features; G fixes the physical quadratic energy and B fits its 10 coefficients. Thus GR and GC fit 35 real parameters, and BR and BC fit 45, using the fixed chart (3). Within each prior choice, both representations use identical function classes, data and regularizers in common real coordinates.

For each seed in $\{17, 43, 91\}$, the two-mode Galerkin model generates 12 conservative 4 s trajectories with 129 observations each. States are noiseless; vector-field components receive uniform noise in $[-10^{-6}, 10^{-6}]$. Regularized least squares fits the readout, and eight independently initialized 16 s trajectories are held out in full. Supplementary Section S5.1 specifies initial-state sampling, regularization and solver tolerances.

Table 3. Five paired contrasts of tip RMSE, using the same eight held-out trajectories for each of three training seeds. Positive prior contrasts favour G; positive representation contrasts favour C.

Contrast	Mean ± sample SD (mm)
R − GR	$(1.08\pm1.44)10^{-5}$
C − GC	$(1.08\pm1.44)10^{-5}$
R − C	$(-4.24\pm4.84)10^{-13}$
GR − GC	$(0.75\pm1.99)10^{-15}$
(GR − GC) − (R − C)	$(4.25\pm4.82)10^{-13}$

Abbreviating BR and BC as R and C, R−GR and C−GC measure the prior effect, R−C and GR−GC compare representations, and (GR−GC)−(R−C) measures their interaction. Differences are paired within each training seed before computing means and sample standard deviations.

The mean tip RMSE is 2.28×10^{-5} mm for BR/BC and 1.20×10^{-5} mm for GR/GC. The prior improves two seeds and worsens the third. Representation and interaction contrasts are smaller than the reference-solver refinement change, 1.92×10^{-8} mm; the matched real and complex fits are numerically equivalent here. The three-seed summaries in Table 3 are descriptive. GC estimates $a = 1.00326 \pm 0.00154$ with nonzero training residuals. Supplementary Table S4 gives slope and physical target-energy errors as well as tip errors. The small errors reflect an in-class target and accurately controlled synthetic data.

A separate 200 s continuation spans 363 first-mode periods and gives GC tip RMSEs of 1.38×10^{-4}–4.03×10^{-4} mm, maximum modal phase error 4.19×10^{-4} rad, and own-energy relative drift below 5.11×10^{-9}. These DOP853 trajectories test long-time prediction, independently of the Central4 test (supplementary Figure S3). Their finite-time agreement leads to the separate invariant-torus question in Section 4.

3.4 Dissipation and independent finite-element comparisons

Driven tests first isolate the role of dissipation. Fitting the Rayleigh coefficient on unforced trajectories gives tip RMSEs of 6.40×10^{-5}–1.37×10^{-4} mm on six held-out forced trajectories; omitting it gives approximately 3.01 mm. The largest work-balance residual with dissipation is 4.41×10^{-17} J. These free-motion tests precede the clearance constraints introduced in Section 5.

Independent COMSOL calculations then quantify reduction error. On an 8 s driven test, exact-cosine predictors with 2, 4 and 8 modes give tip RMSEs of 0.0156168, 0.00407422 and 0.00102484 mm, respectively, versus 0.00005614 mm for all 64 modes. The learned two-mode GC error,

0.0156171–$0.0156177\,\mathrm{mm}$, shows that reduction dominates this test. Supplementary Section S5.4 gives the assembly and refinement checks.

Fitting position–velocity increments from separate full-FE COMSOL trajectories gives $a = 1.0223248822$ and test errors 0.14836 and $0.05140\,\mu\mathrm{m}$, versus 0.24172 and $0.27459\,\mu\mathrm{m}$ for the exact two-mode predictor. The learned residual partly absorbs reduction error (Supplementary Section S5.5). The KAM proof instead uses separate two-mode COMSOL observations, so that its identification error can be bounded directly.

Unknown-parameter and model-mismatch tests further assess the feature choice (Supplementary Section S5.7). For a soft rod with positive field gradient, adding omitted physical features reduces tip RMSE from 182.988 to $9.829\,\mu\mathrm{m}$; a freely fitted quadratic model gives $7.258\,\mu\mathrm{m}$. At 0.5% state noise, the corresponding errors are 183.028, 9.757 and $7.216\,\mu\mathrm{m}$. Thus a physical prior can limit absorption of reduction error. Finally, on six one-second forecasts along a slowly drifting full-FE trajectory, causal adaptation from past states reduces mean error from 387.483 to $9.867\,\mu\mathrm{m}$. These nonautonomous forecasts have a separate empirical scope from the stationary KAM result.

4 Connection to KAM nondegeneracy and invariant tori

4.1 From state observations to analytic error

Long-time prediction tests the fitted dynamics along selected trajectories. To establish invariant motion, we also need error bounds on a neighborhood of those trajectories. Following the learning-to-KAM connection of Chen et al. (2022, Theorem 11), we derive these bounds for the analytic readout (20), $h_\eta = h_2 + \eta^{\mathsf{T}} F$, whose quadratic part is fixed and whose full-cosine target is $h_\star = h_{\eta_\star}$. We write $\widehat{h}$ for the fitted function, including coefficient-export rounding.

For independently sampled gradient labels, our uniform learning bound in Supplementary Theorem S3 connects nonzero loss to a C^ℓ error under explicit sampling and regularity assumptions. Supplementary Corollary S4 combines it with Bounemoura (2020, Theorem A). The COMSOL observations used here are correlated states along trajectories, so we instead derive a deterministic bound from their integrated equations of motion.

COMSOL supplies 17 position–velocity observations on each of $m = 64$ conservative two-mode trajectories. Let $c_j : [0, T] \to \mathbb{R}^4$ be their degree-eight fitted curves, with endpoints fixed to the observations, and let $Lx = J\nabla h_2(x)$. Define

$$
\begin{aligned}
D_j &= T^{-1} \int_0^T J\nabla F(c_j(t))\, dt, \\
y_j &= T^{-1} \left[c_j(T) - c_j(0) - \int_0^T L c_j(t)\, dt \right], \\
G_I &= (4m)^{-1} \sum_j D_j^{\mathsf{T}} D_j.
\end{aligned}
\tag{22}
$$

These labels require states and feature integrals, without acceleration reconstruction. Let r_I be the RMS residual $D_j\widehat{\eta} - y_j$ over the $4m$ scalar observations, and let ϵ_I bound the RMS integrated physical defect $e_j = T^{-1}\int_0^T (\dot{c}_j - J\nabla h_\star(c_j))\, dt$.

Proposition 1. Integral observations bound the learned energy. *If* $\lambda_{\min}(G_I) \geq \underline{\lambda} > 0$, *then*

$$
\begin{aligned}
\|\widehat{\eta} - \eta_\star\|_2 &\leq b_I := \frac{r_I + \epsilon_I}{\sqrt{\underline{\lambda}}}, \\
\|\widehat{h} - h_\star\|_r &\leq b_I B_F(r) + \delta_{h,\mathrm{dec}}(r).
\end{aligned}
\tag{23}
$$

Here $\|\cdot\|_r$ *is the supremum norm on the complex polydisc* $|x|_\infty < r$, $B_F(r)^2 = \sum_j \|F_j\|_r^2$, *and* $\delta_{h,\mathrm{dec}}$ *bounds coefficient-export rounding.*

Proof. Endpoint matching gives $y_j - D_j\eta_\star = e_j$. The triangle inequality in the normalized stacked observation norm bounds $\sqrt{\underline{\lambda}}\|\widehat{\eta} - \eta_\star\|_2$ by $r_I + \epsilon_I$. Cauchy–Schwarz in feature space then gives the analytic bound.

The physical defect is evaluated against the declared target; it is additional information beyond the training loss. Interval integration bounds this defect and retains the full sine remainder and coefficient rounding. Cauchy's estimate then bounds every derivative on a smaller polydisc. Supplementary Section S4.5 gives these estimates and the independent trajectory-error bound.

4.2 The magnetic nonlinearity preserves twist after learning

The analytic error bound becomes useful for KAM theory because the magnetic potential supplies a nondegenerate frequency map. For $h_\star = h_2 + g(d^{\mathsf{T}}X)$, with $g(u) = 1 - \cos u - u^2/2$, its quartic part is $-(d^{\mathsf{T}}X)^4/24$. Averaging over the modal angles with $I_j = (X_j^2 + Y_j^2)/2$ gives the action Hessian

$$
\begin{aligned}
A_\star &= -\frac{1}{8}\begin{pmatrix} d_1^4 & 2d_1^2 d_2^2 \\ 2d_1^2 d_2^2 & d_2^4 \end{pmatrix}, \\
\det A_\star &= -\frac{3}{64} d_1^4 d_2^4 \neq 0.
\end{aligned}
\tag{24}
$$

Table 4. Learning error and twist margin for the COMSOL two-mode fit. All displayed bounds include coefficient rounding and are rounded outward.

Quantity	Bound
Training MSE	$[3.6030, 3.6031]\,10^{-23}$
r_I, ϵ_I	$\le 6.003\,10^{-12}$, $6.884\,10^{-12}$
$\lambda_{\min}(G_I)$	$\ge 2.930\,10^{-7}$
b_I	$\le 2.381\,10^{-8}$
Twist error	$\le 1.038\,10^{-7}$
$\sigma_{\min}(A_\star)$	$\ge 4.903\,10^{-4}$
$\|\widehat{h} - h_\star\|_{0.3}$	$\le 1.331\,10^{-9}$

Indeed, $\langle X_j^4\rangle = 3I_j^2/2$ and $\langle X_1^2 X_2^2\rangle = I_1 I_2$, while odd terms average to zero. Both modal tip rotations are nonzero, so the twist is nonsingular, although indefinite.

Learning preserves this property if its quartic error is smaller than the target's least singular value. Let B map the readout coefficients to quartic coefficients and let δ_{dec} bound rounding in that map. Angular averaging yields

$$\|A_{\widehat{h}} - A_\star\|_2 \le \sqrt{19}\big(\|B\|_2 b_I + \delta_{\mathrm{dec}}\big) < \sigma_{\min}(A_\star). \quad (25)$$

The factor $\sqrt{19}$ bounds the quartic-to-twist operator, including momentum-dependent terms. Table 4 establishes the strict inequality for the Hamiltonian learned from the COMSOL states.

The complete magnetic potential also contributes a sixth-order analytic tail bounded by $(Dr)^6\cosh(Dr)/720$, where $D = \|d\|_1$. Together with fourth-order nonresonance, the twist bound gives a local normal form $\nu\cdot I + \frac{1}{2} I^{\mathsf{T}} A_{\widehat{h}} I + O(r^6)$. Our remainder estimates in Supplementary Sections S4.1–S4.2 therefore yield positive-measure KAM families near equilibrium. To establish persistence at a specified physical amplitude, we next apply an a posteriori theorem to the same fitted function.

4.3 An invariant torus at finite physical amplitude

The existence theorem used here is Haro and Luque (2019, Theorem 3.1), with the constants of arXiv:1711.05515v1, Appendix B. Our contribution in the following result is the construction and analytic bounds for the learned magnetic model.

Theorem 2. Finite-amplitude persistence for the learned magnetic model. *The fitted Hamiltonian in Table 4, specified by its exact coefficients in Supplementary Section S4.5, admits a real analytic embedded Lagrangian torus* $K_\infty : \mathbb{T}^2 \to \mathbb{R}^4$ *satisfying*

$$J\nabla\widehat{h}(K_\infty) = DK_\infty\varpi. \quad (26)$$

The exact frequency ϖ *is specified in Supplementary Eqs.* (S68)–(S69)*; angles have period one and time is dimensionless. The torus is within* $2.0894\,10^{-41}$ *of the computed Fourier embedding on the complex strip of width* $1/80$*. Its beam reconstruction attains a tip displacement of at least* 1.8977904 *mm and has centreline displacement at most* 1.8977962 *mm at every phase.*

Proof. The bounded continued-fraction coefficients of the frequency ratio give $|k\cdot\varpi| \ge (9/1750)|k|_1^{-1}$ for every nonzero integer k, and hence the weaker Diophantine exponent 3 used in the theorem. On the strip $\rho = 1/40$, 512-bit ball-arithmetic evaluation of the Fourier candidate, including interpolation tails, bounds the analytic invariance residual by $2.7310\,10^{-77}$ and its inverse averaged torsion by 80526. The geometric and derivative estimates in Supplementary Section S4.3 give a residual threshold greater than $8.6086\,10^{-76}$, with smallness quotient below $0.031724 < 1$. The cited theorem then supplies the exact torus and correction bound. Fourier bounds and the Hermite-element convex-hull property give the displacement estimates.

Thus nonzero learning loss, preserved twist and finite-amplitude persistence are established for one fitted Hamiltonian. The result uses the independently simulated two-mode state observations; the full-beam comparisons in Section 3 quantify a separate reduction error. Section 5 connects this conservative motion to the controlled full beam.

4.4 From one torus to a family of quasiperiodic motions

The nondegeneracy bound survives the a posteriori iteration: $\|\langle T_\infty\rangle^{-1}\| \le 2\,10^5$ for the exact torus's averaged torsion. In an analytic symplectic tube, solving one quadratic homological equation therefore gives

$$\widehat{h}\circ\Psi = e + \varpi\cdot I + \tfrac{1}{2} I^{\mathsf{T}}\overline{A} I + O(|I|^3), \qquad \det\overline{A} \ne 0. \quad (27)$$

Our construction identifies $\overline{A}$ with the averaged torsion and retains an analytic remainder (Supplementary Corollary S8). The nonresonant frequency, nondegenerate quadratic part and analytic remainder meet the hypotheses of Bounemoura (2016, Theorem 6, $\alpha = 1$, arXiv:1412.0509v1). That theorem gives positive-measure families of analytic Lagrangian invariant tori around K_∞, with relative density tending to one as the transverse action box shrinks. The same argument applies to all seven fitted or target Hamiltonians considered in the supplement.

A numerical fraction additionally depends on the chosen initial-state domain. For $U \subset \mathbb{R}^4$ of positive Liouville volume, define $f_{\mathrm{KAM}}(U) = \mu_L(U \cap \mathcal{K}_h)/\mu_L(U)$, where $\mathcal{K}_h$ is the union of analytic Diophantine Lagrangian tori. Supplementary Section S11 applies Chierchia and Koudjinan (2021, Theorems 1 and 3) to an explicit tube; its very small transverse width limits its use as an operating tolerance. For the broader action domain $I_1, I_2 \in [0.0032, 0.0072]$, independent uniform sampling of actions and phases gives $1024/1024$ finite-time quasiperiodicity candidates over 8192 dimensionless time units (717.5 s). At step $1/64$, all pass the eight-digit test on consecutive windows of length 2048 and 4096; six- and ten-digit tests agree. Supplementary Section S5.6 gives the weighted Birkhoff method, uncertainty and refinement checks. This broader-domain diagnostic complements the local family theorem without supplying an infinite-time volume bound. The magnetic realization below carries the family's low-mode parameter measure into the damped plant; this measure is distinct from volume in the full 128-dimensional state space.

5 Guidance for magnetic actuation and clearance

The invariant torus established in Section 4 specifies a bounded conservative motion. We now derive the magnetic commands that sustain it in the damped beam and bound the response of all remaining modes. This construction gives an operating envelope for recurrent motion. A physical-energy constraint then extends clearance control to prescribed endpoint manoeuvres.

5.1 Realizing the learned dynamics through two magnetic ports

Let a_j, v_j denote the mass-normalized positions and velocities of all 64 transverse finite-element modes. Their bias-field frequencies are ω_j, and their tip displacement and rotation coefficients are b_j, t_j. A transverse tip force F and a transverse field B_y give

$$\begin{aligned} \ddot{a}_j + \alpha \dot{a}_j + \omega_j^2 a_j &= b_j F + t_j U, \\ U &= E_0(\vartheta - \sin\vartheta) + \mu B_y \cos\vartheta, \\ \vartheta &= \sum_j t_j a_j. \end{aligned} \tag{28}$$

These complete modal coefficients define the plant used below; Supplement S6.1 gives its relation to the assembled finite-element matrices.

For the first two modes, write $a_i = S_i X_i$ with the diagonal scales of Section 2. Because the learned h contains momentum quartics, velocity determines Y through the implicit relation $h_Y(X, Y) = \mathrm{diag}(S_i)^{-1} v_{1:2}/\omega_0$. If $h_{YY} \succ 0$, this relation is locally invertible. Differentiating it along $\dot{X} = \omega_0 h_Y$, $\dot{Y} = -\omega_0 h_X$ gives the desired physical acceleration and hence the feedback

$$\begin{aligned} a^{\mathrm{des}}_{i,tt} &= S_i \omega_0^2 (h_{YX} h_Y - h_{YY} h_X)_i, \\ \begin{pmatrix} F \\ U \end{pmatrix} &= A_{\mathrm{mag}}^{-1} \left(a^{\mathrm{des}}_{1:2,tt} + \alpha v_{1:2} + \mathrm{diag}(\omega_1^2, \omega_2^2) a_{1:2} \right), \\ A_{\mathrm{mag}} &= \begin{pmatrix} b_1 & t_1 \\ b_2 & t_2 \end{pmatrix}. \end{aligned} \tag{29}$$

Substitution into (28) enforces the desired acceleration. Differentiating the velocity relation once more and using invertibility of h_{YY} then recovers both canonical equations.

The force and moment in (29) are physically realized by the local field $B(r) = (B_0, B_y) + G(r - r_{\mathrm{tip}})$, where

$$\begin{aligned} B_y &= \frac{U - E_0(\vartheta - \sin\vartheta)}{\mu \cos\vartheta}, \\ G &= \frac{F}{\mu} \begin{pmatrix} -\sin\vartheta & \cos\vartheta \\ \cos\vartheta & \sin\vartheta \end{pmatrix}. \end{aligned} \tag{30}$$

For the dipole $m = \mu(\cos\vartheta, \sin\vartheta)$, direct multiplication gives $Gm = (0, F)$ and $\|G\|_2 = |F|/\mu$. Symmetry and zero trace of G satisfy the local source-free curl and divergence constraints. The construction requires independent field and gradient authority, observed modal and tip states, and $|\vartheta| < \pi/2$; it does not assume independently actuated mechanical modes.

Theorem 3. Full-model realization of the certified torus. *Suppose the learned Hamiltonian has an analytic embedded two-torus with physical angular frequency $\Omega = 2\pi\omega_0\varpi$. Assume $h_{YY} \succ 0$ on a domain containing this torus with convex Y-fibres for each X, A_{mag} is invertible and $\alpha > 0$. If the lifted motion stays in the domain $|\vartheta| < \pi/2$ of the available commands, feedback (29)–(30) gives an embedded invariant two-torus of the complete finite-dimensional plant. For each low-mode orbit on that torus, higher-mode deviations from its unique quasiperiodic response decay exponentially along closed-loop trajectories that remain in the stated command domain.*

Proof. The feedback reproduces the low-mode Hamiltonian exactly. Its demands F, U therefore depend only on the torus phase. With Fourier convention $e^{ik\cdot\phi}$, each higher-mode

Table 5. Rigorous bounds for the COMSOL-integral learned model. Startup means zero higher-mode positions and velocities, with the low state on the certified torus.

Quantity	Upper bound
Centreline on response graph	1.922696 mm
Centreline including startup	1.970263 mm
Transverse field including startup	0.921665 mT
Gradient norm including startup	0.02572538 T/m

response is

$$a^*_{j,k} = \frac{b_j F_k + t_j U_k}{\omega_j^2 - (k\cdot\Omega)^2 + i\alpha(k\cdot\Omega)}, \qquad j \geq 3. \tag{31}$$

The denominator is nonzero and its inverse is bounded on real frequencies: at zero its real part is positive, and away from zero its imaginary part is nonzero. Analytic forcing thus gives an analytic response on a possibly smaller strip. Positive definiteness on each convex Y-fibre makes $Y \mapsto h_Y(X,Y)$ strictly monotone and injective. The physical low-mode coordinates therefore preserve the torus embedding. Differences between high-mode solutions satisfy homogeneous damped oscillator equations and decay exponentially.

This lift is derived here from the model and the torus supplied by the external KAM theorem. Its physical angular frequency is approximately $(11.41719351, 70.24034116)$ rad/s. The positive-measure family remains a statement about the four-dimensional low subsystem. Without sustained input, the physical beam instead converges to its globally asymptotically stable equilibrium (Supplementary Proposition S2).

For the COMSOL integral fit, interval bounds give $h_{YY} \succeq 0.9999999983I$ on the real 0.3 box and exclude zero from $\det A_{\mathrm{mag}}$. The response and startup estimates in Supplement S6.2 include all 62 higher modes and the complete nonlinear forcing remainder. Cubic-Hermite spatial bounds yield the whole-centreline and command limits in Table 5. With the low state on the exact certified torus and all higher positions and velocities initially zero, the slope remains below 0.155261 rad. Hence $\cos\vartheta > 0.987971$, closing the feedback-domain condition for every phase and all time.

Figure 1 shows the full-state realization. Over 12 s, the low-mode state agrees with an independent Hamiltonian integration to within 1.54×10^{-10} in canonical coordinates. Online COMSOL feedback provides a second implementation check: refining its fixed step from 1 to 0.125 ms reduces the four-second tip RMSE against the independent full-state solver from 147.852 to 3.848 μm. The analytic and interval estimates supply the all-time bounds in Table 5; Supplement S6.1 reports the complete refinement study.

5.2 Frequency continuation and the clearance boundary

To test where driven motion changes character, we vary a single sinusoidal transverse field from 1 to 20 mT and its frequency ratio $r = \omega_f/\omega_0$ from 0.7 to 1.3. The two-mode physical model is evaluated at 1,029 independently initialized points and 686 upward/downward continuation points. Each run discards 128 forcing periods and diagnoses the next 64 periods. All tested responses converge to forcing-periodic motion; their numerical Floquet spectral radii range from 0.803809 to 0.889055. These observations locate no quasiperiodic or chaotic transition within the sampled protocol. Large-deflection extrapolations are separately flagged, and this finite scan does not establish global absence of chaos or breakup of the conservative KAM tori.

A distinct practical boundary appears when periodic bending reaches the 4 mm clearance. Solving for periodic responses with all 64 modes gives threshold amplitudes of approximately 7.5005, 1.91685 and 8.7909 mT at $r = 0.9$, 1 and 1.1, respectively. These are numerical steady-response thresholds; the specified startup ramp can cross the clearance earlier. Supplement S5.9 reports the initial states, continuation protocol, transient thresholds, modal and temporal refinement, and independent full-model checks. The resonance boundary motivates a work constraint that covers transients as well as settled motion.

5.3 Clearance beyond the invariant torus

Tracking a prescribed path need not keep the state on a conservative torus. We therefore derive a clearance condition from the physical energy itself. For the clamped small-slope beam, $w(s) = \int_0^s (s-\xi)w_{ss}(\xi)\,d\xi$. Since the static magnetic energy is nonnegative, Cauchy–Schwarz gives

$$|w(s,t)|^2 \leq \frac{2s^3E(t)}{3EI}, \qquad E_c = \frac{3EIc^2}{2L^3}, \tag{32}$$

where c is the corridor half-width less the rod radius. Thus $E < E_c$ guarantees clearance of the entire centreline and its radius tube. This argument applies to sufficiently regular continuum solutions satisfying the power identity and separately to conforming Hermite finite-element solutions.

For a single transverse magnetic input, its actual power is a_pB_y, with $a_p = \mu\cos\vartheta\,\dot\vartheta$. Starting from a valid bound $\overline{E}_0 \geq E(0)$, let $\dot W = a_pB_y$ and $W(0) = \overline{E}_0$. Nonnegative dissipation gives $E \leq W$. Choose $\overline{E}_0 < E_{\max} < E_c$, and

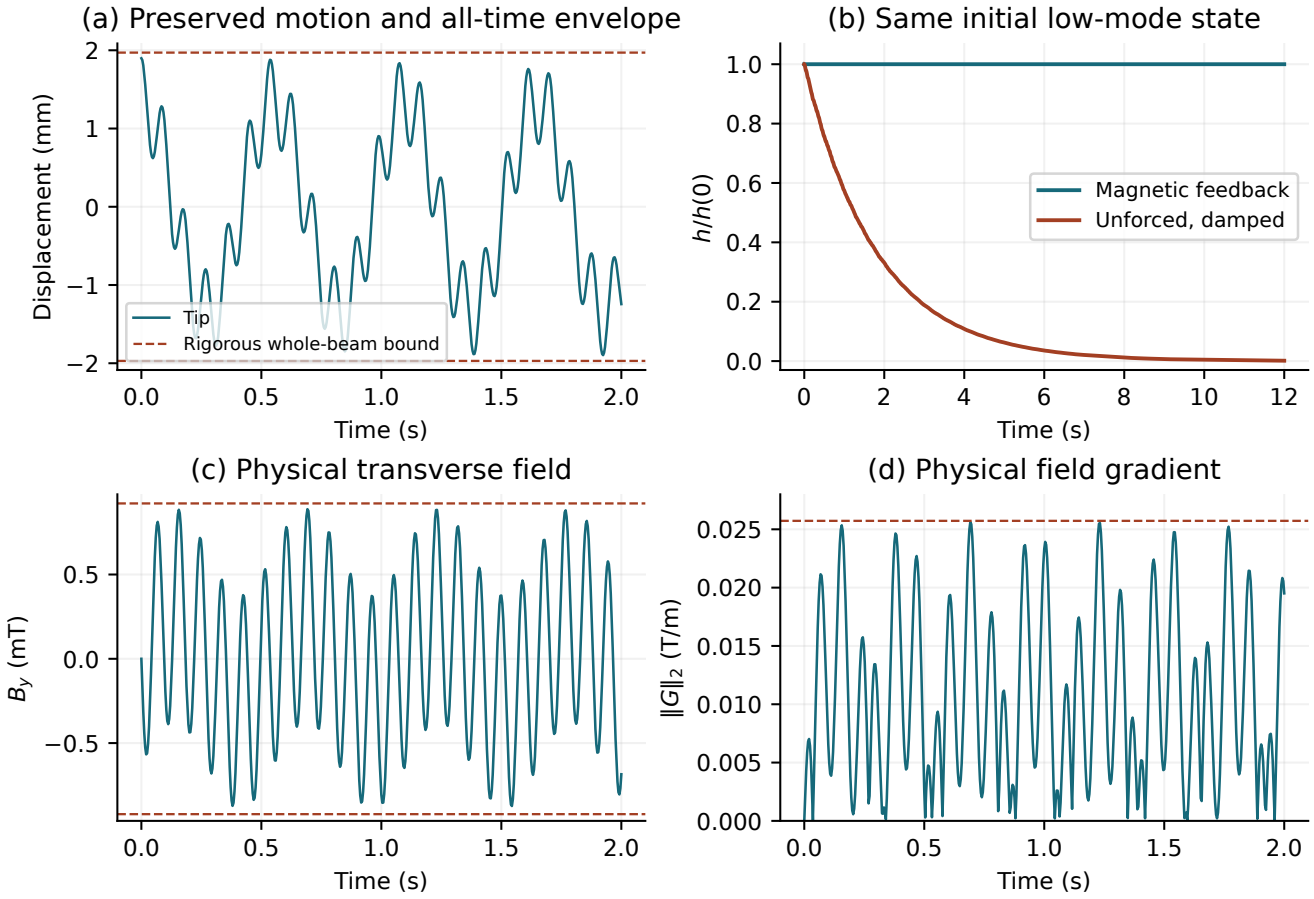


Figure 1. Realizing the certified learned dynamics on all 64 modes. The higher modes start at zero. Dashed envelopes include the startup transient and bound every phase and centreline point. Magnetic field and gradient jointly supply the force and torque; the passive comparison uses the same initial state without this feedback. Panels (a,c,d) show 2 s and panel (b) shows 12 s.

project the requested field onto

$$|B_y| \le B_{\max}, \qquad a_p B_y \le \kappa(E_{\max} - W), \quad \kappa > 0. \tag{33}$$

Zero field is feasible while $W \le E_{\max}$. Moreover, (33) implies $E_{\max} - W(t) \ge (E_{\max} - \overline{E}_0)e^{-\kappa t}$. Together with (32), this proves all-time clearance under exact port measurement and command application. The result applies to driven trajectories through their supplied work, including motions away from a KAM torus; additional work ports or measurement errors must enter the same budget.

For a 4 mm net clearance, $E_c = 23.8565$ nJ. An energy budget $E_{\max} = 0.95E_c$ bounds the centreline by 3.898718 mm in continuous time. Under a 10 mT sinusoidal request, the sampled peak falls from 5.14478 mm without the governor to 3.14362 mm with it; online COMSOL gives 3.143643 mm. Supplement S7.1 supplies the power derivation, an open-loop field-variation bound and refinement data.

5.4 Nonzero sensing, initialization and actuation tolerances

The extremely narrow action tube in the KAM measure estimate is a proof domain, rather than an initialization tolerance. A separate physical-energy argument yields a nonzero operating tolerance about the exported finite Fourier reference. In the frozen full nodal model, let $Q(e) = \frac{1}{2}e_v^{\mathsf{T}} M e_v + \frac{1}{2}e_q^{\mathsf{T}} K_b e_q$. The initialization set $Q(e) \le 5$ pJ implies a tip-position error below 56.006 μm; this is a projection of an energy ellipsoid, not a box of independently admissible coordinate errors.

For the two work ports, bound tip-angle, angular-rate and tip-velocity measurement errors by 1 mrad, 0.01 rad/s and 0.1 mm/s. Allow 1% multiplicative error in both applied inputs, together with additive errors of 0.5 nN in force and 0.05 μT in field. A worst-case measured-power bound and a dissipation-based reserve of 0.096780 nJ cover these additive errors over 4 s. The resulting 22.66 nJ energy cap bounds the whole centreline below 3.898405 mm throughout that interval. Supplement S7.5 gives the proof, directed-rounding certificate, actuator derating and full-model refinement. The governor can alter the requested motion substantially: this

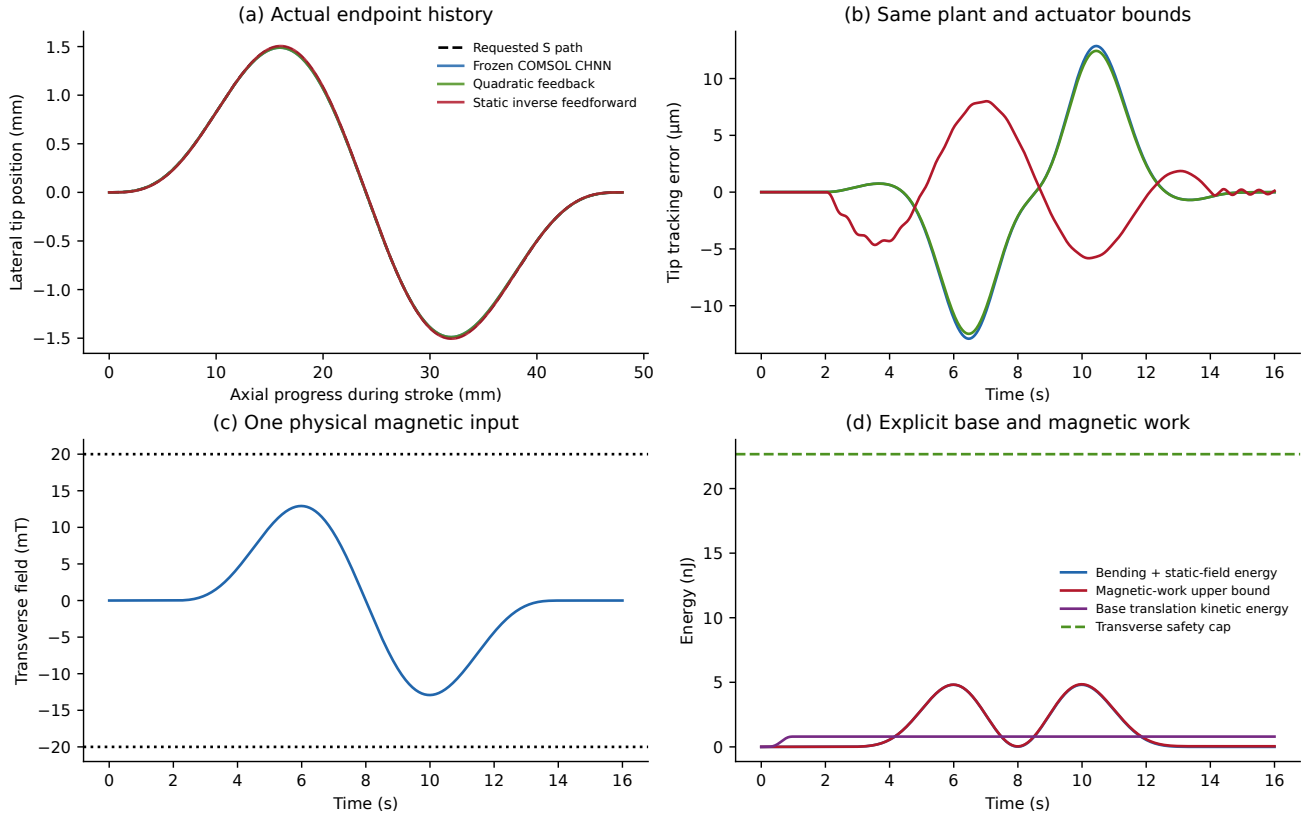


Figure 2. S-shaped endpoint tracking through axial base feed and transverse magnetic actuation. The same full beam and input limits are used for the matched controls. Base work is included in the energy balance.

guarantee concerns clearance under the stated errors and does not preserve exact torus membership or certify tracking accuracy.

5.5 Endpoint tracking and practical scope

We next combine axial base translation with magnetic bending to trace an S-shaped endpoint path (Figure 2). The tip advances 48 mm over a 12 s stroke with lateral extrema ± 1.5 mm. Axial acceleration ends before bending begins, and the model has no axial drag or guide contact. With base kinetic energy and work accounted for separately, the transverse governor bounds clearance in the straight corridor (Supplement S7.6). The prescribed S shape is the moving endpoint trajectory.

The controller uses the two-mode Hamiltonian fitted to full-FE COMSOL trajectories, its full gradient and Hessian, observed low-mode projections and measured tip feedback, with coefficients fixed throughout the task. Table 6 compares controllers on the same trajectory and plant with a 20 mT field limit. The learned controller achieves $6.419\,\mu$m RMS error, while the known-physics inverse performs better on this slow nominal path. The modal-output comparison shows the benefit of using the measured tip in feedback.

Table 6. Matched S-path controls. RMS errors use the 12 s stroke; maxima use the complete 16 s simulation.

Control	RMS (μm)	Max. (μm)
CHNN, tip feedback	6.419	12.907
Quadratic, tip feedback	6.207	12.476
Static inverse feedforward	4.124	8.005
CHNN, modal-output feedback	25.308	41.544

The learned command peaks at 12.9177 mT and the centreline at 1.48892 mm. The maximum recorded work-based energy estimate is 4.84997 nJ, and the governor remains inactive for this request. An independent COMSOL replay of the field agrees with the predicted tip to within 4.038×10^{-10} m. The replay tests this task's mechanical response; the torus experiment above additionally tests online feedback.

The resulting design guidance is to preserve fitted energy derivatives, check actuation rank and momentum invertibility, and budget all input work. The envelopes require the stated state estimates and local field/gradient authority. Device implementation requires calibrated hardware and verified error bounds; partial sensing, large rotations, flow and wall contact need corresponding extensions.

6 Conclusion

We connected an identified magnetic energy to numerical prediction, nonlinear persistence and physical actuation in four steps. The complex representation and classical central symplectic update preserve the model's structure; the approximation analysis carries nonzero identification error into the KAM hypotheses; and magnetic feedback realizes the resulting finite-amplitude torus in the damped full beam. The model-specific realization and lift yield explicit centreline, field and gradient bounds. Input-work control extends clearance guarantees to driven motions beyond that torus, with explicit nonzero initialization, sensing and actuation tolerances over a prescribed finite horizon. The frequency study identifies a resonance-induced clearance boundary; no chaotic transition is detected under its sampled protocol.

The numerical comparisons define the scope of this result: real and complex implementations have equivalent resolved accuracy, and the known-physics inverse performs best on the slow nominal tracking task. Extending the analysis to partial sensing, calibrated actuators and three-dimensional contact requires corresponding mechanical models and error bounds.

Data and code availability

The supplied materials comprise two identified archives. Archive S1 is the complete computational record dated 2 October 2026, containing native COMSOL models, construction and learning code, trajectories, fitted coefficients and interval records. Archive S2, `S2_citation_revision_20261008.zip`, contains the revised article and Supplement, the additional stability and tolerance studies, frozen inputs, raw numerical outputs and reproduction commands. The accompanying `reproducibility/ARTIFACT_INDEX.md` identifies the archives and maps results to files and checksums; its source map separates full-FE identification from the two-mode integral data used for the KAM proof.

Declarations

This work reports mathematical analysis and computer-generated mechanical data. No human, animal or in-orbit experiment was performed.

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